

SECTION 11555-C
CDA 離心式空壓機採購規範

1.0 總則

- A. 設置地點：Micron Falcon CDA Room
- B. 交 期：PO 後 6 個月內到廠
- C. 空壓機型式：離心式空壓機
- D. 數量：4 台
- E. 規格：(詳如下述)

1.1 本設備包含無油三段離心式空氣壓縮機、中間冷卻器、後部冷卻器、潤滑油系統、控制盤、6600V 降壓啟動盤與馬達等，啟動盤測須第三公正單位(電機技師簽證)完成，啟動盤內重要設備須與業主討論同時須要業主同時。

- A. 空壓機之設計條件為：
 - 1. 大氣壓力： 1.033 kg/cm^2 (A)
 - 2. 入口壓力： 1.012 kg/cm^2 (A)
 - 3. 入口空氣溫度： 40°C
 - 4. 周圍相對濕度：90 %
 - 5. 冰水溫度： $12^\circ\text{C}/20^\circ\text{C}$ ，溫水： $29^\circ\text{C}/35^\circ\text{C}$
- B. 空壓機之出氣壓力及出氣風量須高於以下所列：
 - 1. 出口壓力：設計操作壓力 9.5 kg/cm^2 (G)。
 - 2. 最大操作壓力 10.5 kg/cm^2 (G)以上。
 - 3. 出口風量： $6000 \text{ NCMH} / 9.5 \text{ kg/cm}^2$ (G) (N: normal 0°C 1atm RH: 0%) 以上。
 - 4. 以上資料需附性能曲線。

1.2 冷卻方式：

- A. 以潤滑油冷卻壓縮機本體，以溫水冷卻中間冷卻器/後部冷卻器，冷卻水控制閥與運轉連鎖，運轉及待機時均不得有結露之現象。
- B. 熱交換管材質為銅管鋁鰭片。

1.3 空壓機需能 24 小時連續滿載運轉。

1.4 壓縮機空氣進口需含兩段濾網(可拆式前/後空氣濾清器)及消音器，空氣濾網需考慮如何防止異物進入損壞葉片。

1.5 空氣出口需附有止回閥。

1.6 壓縮段需有振動保護。

1.7 運轉控制模式

- A. Load/Unload control (空車、重車控制)。
- B. Constant Pressure Control (恆壓控制)。
- C. Auto-Dual Control (自動雙重控制)，需標示可控制範圍。
- D. 進口容調控制結合卸載控制 (節能控制)
- E. 自動起停控制 (Auto-Start/Stop Control)
- F. Turn Down 70% ~ 100% at 10.5 kg/cm² (G)。

1.8 操作控制盤：每一台空壓機內建一只電腦操作控制盤。

A. 操作控制盤需具備下列操作功能：

- 1. 10" 以上觸控顯示操作面板
- 2. 啟動 / 停機鍵。
- 3. 警示功能。
- 4. 功能復歸。
- 5. 運轉故障自動記錄(200 筆以上)。
- 6. PLC 控制，停電後復電自動起機 (15sec 內)。
- 7. 緊急停機按鈕 (附防撞保護)。

B. 操作控制盤需具備下列顯示功能及顯示裝置：

- 1. 出口壓縮段排氣壓力。
- 2. 出口壓縮段排氣溫度。
- 3. 系統設定壓力。
- 4. 馬達運轉電流。
- 5. 運轉時間積算。
- 6. 電壓供應是否正常。
- 7. 進/排氣閥開度指示。
- 8. 空濾芯壓差顯示警示
- 9. 油濾芯壓差顯示警示

C. 需具備下列保護功能：

- 1. 氣封低壓、低油壓及低油溫時空壓機自動自鎖，防止不正常情況下啟動。
- 2. 油壓過低警示及自動停機功能。
- 3. 油溫過低或過高警示及自動停機功能。
- 4. 各段排氣溫度過高警示及自動停機功能。
- 5. 各壓縮段高振動警示及自動停機功能。
- 6. 氣封氣壓過低自動停機功能。

- 7. 空壓機突破警示及自動卸載功能。
- 8. 電壓供應三相不平衡警示及自動停機功能。

- D. 控制箱體應符合 NEMA4 構造，同時固定於壓縮機座上，控制箱體儀控管路(IA)以不鏽鋼管配裝，所有線路及管路均應在製造廠內完成並測試。
- E. 控制箱體須具備良好散熱能力 ($<45^{\circ}\text{C}$)。
- F. 控制盤之功能顯示、單位需為公制。
- G. 動力幹線須明確標示。

1.9 中間冷卻器及後部冷卻器：(需明確標示報價設備材質)

- A. 內部需作防鏽處理。
- B. 冷卻方式為水走管內，氣走管外式設計。
- C. 各段冷卻器附 BEKO 電子無壓式排水器

1.10 潤滑油系統

- A. 儲油槽內需加以防鏽處理。
- B. 附油加熱器，需有油溫控制。
- C. 主軸驅動全量油泵與電動輔助全量油泵皆需要。

1.11 馬達

- A. 馬達額定馬力：供應商建議。
- B. 電源：6600V，3Ph，60Hz。
- C. Service Factor：1.15。
- D. 附內部加熱器。
- E. R.V. 電抗啟動 Tap：65% & 100%。
- F. 附馬達啟動盤組。
- G. 馬達需試車完成並達到相關性能後始得出廠交貨。（附證明文件）
- H. 馬達絕緣等級：F
- I. 馬達須為歐、美、日、台灣。
- J. 馬達防護等級：T.E.F.C。
- K. 空壓機馬達的軸承須為 sleeve bearing(套筒軸承), 加油方式為油 pump 強力加油作潤滑用，而非一般黃油(grease)式。

1.12 以上資料請依業主提供之評估表格，明確表示報價設備規格材質，於議價前進行規格澄清，若拒作或未交付業主審查者，視同棄權論。

2.0 試車

- A. 供應廠商須配合協助試車。
- B. 各種控制模式測試。
- C. 動平衡測試。
- D. 功能測試：連續運轉 24 小時，同時進行各種校正，設定與調整。
- E. 廠商需供應之範圍包括：
 - 1. 設備與附件的測試及調整工作。
 - 2. 提供控制用 Sensor 校正報告。
 - 3. 配合工程承包商進行空壓機安裝開車調整。
 - 4. 空壓機原廠試車，其試車條件與程序需符合 ASME 標準。
 - 5. 提供工程承包商下列資料
 - a. 空壓機尺寸及重量（含維修空間）。
 - b. 銜接點尺寸及規格。
 - c. 控制盤內所有控制點及監視點須無條件全部開放，以利監控整合。
 - d. 提供技術支援，使設備正常運轉。
 - e. 安裝注意事項。
 - f. 以上資料未提供或提供不確實，致使工程延誤或引發介面糾紛時，設備廠商須負責相關工程修改費用。
 - 6. 交件資料：(3 份，中文&英文)
 - a. 使用操作說明書。
 - b. 維修保養手冊。
 - c. 原廠出廠檢驗證明。
 - d. 提供啟動盤與控制線路圖。
 - e. 提供控制流程圖。
 - f. 史歷卡、SOP 及 Check List。
- F. 報價需附資料：
 - 1. 報價單。
 - 2. 設備型錄、規格說明書、性能曲線、設備及附件之圖面、十年內運轉備品表。
 - 3. 保固貳年並提供貳年份消耗性備品，須於合約中明列名稱及數量。

4. 供應商得標後須提供安裝、操作及維修手冊(中英文)各參份。
 5. 空壓機規格明細 Yes/No Form 表。
- G. 驗收：
供應商協助開車並協助作各項校正、設定與調整、連續運轉 24 小時並作功能之測試（需表列測試項目），合格後製作完整竣工資料一式四份始與驗收。
- H. 叫修服務：
空壓機供應商需提供原廠服務系統組織供評估，並應於叫修時可 4Hr 到廠維修服務。
- I. 操作、保養維修訓練：

空壓機供應商需提供操作維護訓練課程 24 小時以上，訓練業主所指派之操作維護人員，同時分階段執行。
- J. 詳細之規格請參考 YES / NO FORM。
- K. 保固期：

廠商須全責保固貳年，得標廠商二年內須負責所有耗材及保養工資，保固為自驗收日後二年(保養內容須與相同機型相同保養內容與耗材)。
- L. 設備振動：

空壓機運轉(Full Loading)時，其 Overall (3-10K Hz)振動值必須在 5mm/sec 以下。
- M. 空壓機訊號須傳回 FMCS：
1. 設備狀態。
 2. 設備壓力。
 3. 設備水溫。
 4. 電流。
 5. 油溫及油壓。
 6. 出口空氣溫度。

AIR COMPRESSOR SPECIFICATION CHECKLIST

Purchase No.:			Date:	
	AIR COMPRESSOR SPECIFICATION CHECKLIST	VENDOR QUOTATION		
ITEM	SPECIFICATION	YES	NO	EXPLANATION (IF NO)
1	EQUIPMENT: OIL FREE, CENTRIFUGAL TYPE			
2	QUANTITY: 1 SET			
3	LOCATION: INDOOR			
4	PROPOSED MODEL: BY VENDOR			
5	REQUIRED PERFORMANCE:			
	5-1 Capacity: 12000 NM ³ / Hr			
	5-2 Discharge Pressure (POD): 9.5 kg/cm ² (G)			
6	SITE CONDITIONS			
	6-1 Location: TCP2 CDA Room			
	6-2 BAROMETRIC PRESSURE: 1.033 kg/cm ² (A)			
	6-3 Inlet Pressure: 1.012 kg/cm ² (A)			
	6-4 Inlet Temperature: 40°C			
	6-5 Relative Humidity: 90%			
7	COOLING WATER & AIR REQUIRED:			
	7-1 Cooling Water Temperature: inlet 29°C / outlet 35°C & 12°C/20°C			
	7-2 Cooling Water Pressure: 2~4 kg/cm ² (G)			
	7-3 Auxiliary Air Source: 4~7 kg/cm ² (G)			
8	POWER SUPPLY:			
	8-1 Main Power: 6600 V / 3 PH / 60 Hz			
	8-2 Auxiliary Power: 480V / 3 PH / 60 Hz			
	8-3 Oil Heater Power: 480V / ____ PH / 60 Hz			
9	NOISE: Below 90 dBA At 1M From Equipment Distance			
10	MAIN MOTOR:			
	10-1 Motor Rated Horse Power: _____ Pole: _____			
	10-2 Manufacturer: _____			
	10-3 Motor Enclosure: <u>T.E.F.C</u>			
	10-4 Time Rating: <u>Continuous 24 Hours</u>			
	10-5 Mounting Type: _____			
	10-6 Bearing Type: Sleeve bearing forced-lubrication			
	10-7 Rotation: _____			
	10-8 Temperature Rise Limit: 105 °C			
	10-9 Insulation: F Class			
	10-10 Space Heater: 110 V / ____ PH / 60 Hz			
REMARK: 1. Please quote your prices per the listed specification and quality standards, fill this form, and send it to us along with the quotations. 2. If the specifications and the quality standards of your products quoted are the same as those listed in the form, please mark [✓] in the NO column and state detailed specifications to which your company can make in the EXPLANATION column.				

Vendor:

Representative:

Date:

AIR COMPRESSOR SPECIFICATION CHECKLIST

Purchase No.:			Date:	
	AIR COMPRESSOR SPECIFICATION CHECKLIST	VENDOR QUOTATION		
ITEM	SPEICIFCATION	YES	NO	EXPLANATION (IF NO)
	10-11 Starting Type: Reduce Voltage (65% Tap)			
	10-12 Service Factor: 1.15			
11	AUXILIARY PUMP'S MOTOR:			
	11-1 Motor Rated Horse Power: _____ Pole: _____			
	11-2 Motor Enclosure: T.E.F.C.			
	11-3 Insulation: F Class			
12	STARTER PANEL			
	12-1 Manufacture Accordance: CNS, NEMA4, JIS & TPC Standard			
	12-2 Construction:			
	High/ Low voltage self-supporting metal cold switchgear: Basic			
	structure of switchgear is made with 2.3mm high grade finished			
	steel plate weld with 50*50*4 mm L-shape steel angle frame			
	100*50mm channel base. Front door is made of 3.2mm steel			
	plate. Reinforcing is weld in space up on loading requirement.			
	Each enclosure is designed and installed in very strength and			
	Rigidity.			
	12-3 Painting Treatment:			
	Each enclosure is formed by hydraulic pressed, punch holes and			
	Finished in standard size. The assembled cubicle shall be			
	Chemically cleaned and film of phosphate and then paint red			
	Oxide two coats of basic paint, two coats of attached paint. OR			
	Static electricity epoxy resin painting.			
	12-4 Wiring			
	12-4-1 Bus bar in made of 99.9% purity tin-plate copper			
	12-4-2 Laminated plastic nameplates, mounted externally on the			
	cubicle door, shall be provided identifying each cubicle			
	function. Number code shall be marked on the lines.			
	12-4-3 Wire color: Power line – Black 3.5mm above			
	Control wire – Ac: Yellow 2mm; Dc: Blue 2mm			
	Ct: Black 2mm; Pt: Red 2mm			
	12-5 Quality Test			
	12-5-1 Construction inspection			
	12-5-2 Operation test			
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AIR COMPRESSOR SPECIFICATION CHECKLIST

Purchase No.:			Date:	
	AIR COMPRESSOR SPECIFICATION CHECKLIST	VENDOR QUOTATION		
ITEM	SPECIFICATION	YES	NO	EXPLANATION (IF NO)
	12-5-3 Dielectric test			
13	BASIC COMPRESSOR:			
	13-1 Gear Casing Split: Horizontally Split			
	13-1-1 Casing Material: _____			
	13-2 Base plate material: _____			
	13-2-1 Impeller Type: _____ Material: _____			
	13-2-2 Impeller RPM of Each Stage: _____			
	13-2-3 Impeller Diameter of Each Stage: _____			
	13-3 Bull Gear Quality: AGMA 13			
	13-3-1 Bull Gear Material: _____			
	13-3-2 Bull Gear Bearing Type: _____			
	13-4 Shaft Material: _____			
	13-4-1 Pinion Radial Bearing Type: _____			
	13-4-2 Pinion Axial Bearing Type: _____			
	13-5 Pinion Inner Bearing Material: _____			
	13-5-1 Pinion Bearing Housing Material: _____			
	13-6 Diffuser: Vane Type Material: _____			
	13-7 Pinion Bearing Split: _____			
	13-8 Air & Oil Seal Type: _____			
	13-9 Seal Material: _____			
	13-10 Air & Oil Seal Split: _____			
14	AIR SYSTEM:			
	14-1 Inlet Filter: Two Stage & Panel Type			
	14-1-1 Efficiency: 1 st 10 μ , 98%; 2 nd 2 μ , 99.97%			
	14-1-2 Inlet Filter Δ P Indication			
	14-2 Inlet Valve: Inlet Guide Vane (I.G.V) Type			
	14-3 Air Passages: Coated With Corrosion-Resistance			
	14-4 Silencer: Blow-Off Type			
	14-5 Discharge Check Valve Type:			
	14-6 Blow-Off Valve Type:			
	14-7 Expansion Joint: Inlet and Discharge			
14-7-1 Expansion Joint Material: _____				
15	COOLING SYSTEM: Intercoolers & Aftercooler			
REMARK: 1. Please quote your prices per the listed specification and quality standards, fill this form, and send it to us along with the quotations. 2. If the specifications and the quality standards of your products quoted are the same as those listed in the form, please mark [✓] in the NO column and state detailed specifications to which your company can make in the EXPLANATION column.				

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AIR COMPRESSOR SPECIFICATION CHECKLIST

Purchase No.:			Date:	
	AIR COMPRESSOR SPECIFICATION CHECKLIST	VENDOR QUOTATION		
ITEM	SPECIFICATION	YES	NO	EXPLANATION (IF NO)
	15-1 Fully Removable Tube Bundle Design			
	15-2 Water in Tube & straight Through Tube Type			
	15-3 Cooler Tube Diameter: 3/8"			
	15-4 Cooler Design Pressure / Temperature: _____ kg/cm ² / _____ °C			
	15-5 Material: Tube:			
	Fin: Aluminum with Phenolic Coating			
	15-6 Cooler Shell Material:			
	15-7 Intercoolers & Aftercooler with Beko Zero loss trap			
16	LUBRICATION SYSTEM			
	16-1 Main Oil Pump & Auxiliary Pump:			
	16-2 Oil Cooler: Design For 23°C Cooling Water			
	16-3 Oil Filter:			
	16-4 Cooler Design Pressure / Temperature: _____ kg/cm ² / _____			
	16-5 Type of Mounted Into Baseplate:			
	16-6 Gearbox Pressure Indications			
	16-7 Oil Level Gauge			
	16-8 Automatic Oil Temperature Mixing Valve			
	16-9 Oil Heater: Automatic Thermostatically Controlled			
	16-10 System Oil Pressure:			
17	CONTROL PANEL:			
	17-1 Main Controller Type: 10" Display PLC Control & Faceplate			
	17-2 Control Panel Enclosure: NEMA 4			
18	CONTROL MODE & FUNCTION:			
	18-1 Auto – Dual Control			
	18-2 Load – Unload Control			
	18-3 Constant Pressure Control			
	18-4 Manual Start / Stop Control			
	18-5 Auto Start / Stop Control			
	18-6 Remote Common Alarm / Trip			
19	DISPLAY FUNCTIONS:			
	19-1 Motor Driver Current			
	19-2 Vibration for Each Shaft			
	19-3 Oil Pressure & Temperature			
REMARK:	1. Please quote your prices per the listed specification and quality standards, fill this form, and send it to us along with the quotations. 2. If the specifications and the quality standards of your products quoted are the same as those listed in the form, please mark [✓] in the NO column and state detailed specifications to which your company can make in the EXPLANATION column.			

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AIR COMPRESSOR SPECIFICATION CHECKLIST

Purchase No.:			Date:	
	AIR COMPRESSOR SPECIFICATION CHECKLIST	VENDOR QUOTATION		
ITEM	SPECIFICATION	YES	NO	EXPLANATION (IF NO)
	19-4 Compressor Surge			
	19-5 Discharge Air Pressure			
	19-6 Discharge Air Temperature			
	19-7 Unit Ready to Start Status			
	19-8 Auxiliary Pump Status			
	19-9 Compressor Inlet & Discharge Valve Position			
	19-10 "First Out" of alarm & Trip			
20	PROTECTIVE FUNCTIONS			
	20-1 High Discharge Temperature – Alarm & Trip			
	20-2 High Oil Temperature – Alarm & Trip			
	20-3 High Vibration for Each Shaft – Alarm & Trip			
	20-4 Low Oil temperature – Alarm			
	20-5 Low Oil Pressure – Alarm			
	20-6 Compressor Surge – Alarm			
	20-7 Seal Air Pressure of Low – Alarm			
	20-8 PLC Controller Fault – Trip			
	20-9 High Motor Load – Alarm			
	20-10 Transmitters Fault – Trip			
	20-11 Low Oil Level – Alarm			
	20-12 Air Filter Differential Alarm			
	20-13 Oil Filter Differential Alarm			
20-14 Motor Stator / Bearing Temperature Alarm & Trip				
21	TESTING:			
	Mechanical run and design point performance test in accordance with the ASME power test code, PTC-10			
22	DRAWINGS AND DOCUMENTATION TO BE SUBMITTED UPON RECEIPT OF ORDER			
	22-1 General Arrangement Drawing			
	22-2 Process and Instrumentation Design			
	22-3 Panel Out Line Drawing			
	22-4 Pneumatic Schematic			
	22-5 Electrical Schematic			
	22-6 Main Motor Driver Outline Drawing & Data			
	22-7 Installation and Planning Manual			
REMARK: 1. Please quote your prices per the listed specification and quality standards, fill this form, and send it to us along with the quotations. 2. If the specifications and the quality standards of your products quoted are the same as those listed in the form, please mark [✓] in the NO column and state detailed specifications to which your company can make in the EXPLANATION column.				

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AIR COMPRESSOR SPECIFICATION CHECKLIST

Purchase No.:			Date:	
	AIR COMPRESSOR SPECIFICATION CHECKLIST	VENDOR QUOTATION		
ITEM	SPECIFICATION	YES	NO	EXPLANATION (IF NO)
	22-8 Three Copies of Instruction Manual(English & Chinese Version)			
23	SPARE PARTS			
	23-1 1st Stage Filter Element for 2 Years			
	23-2 2nd Stage Filter Element for 2 Year			
	23-3 Oil Filter Element for 2 Year			
	23-4 Oil Mist Element for 2 Year			
24	GUARANTEE			
	24-1 Performance Guarantee			
	The system shall be guaranteed for satisfactory performance at			
25	WARRANTY:			
	25-1 Vendor have to warrant the equipment for free charge			
	operating conditions			
	25-2 Mechanical Guarantee			
	All equipment and component parts shall be guaranteed by			
	the vendor against defective parts improper materials and			
	poor workman ship in accordance with the terms and			
	conditions of the requests for quotation and purchase.			
	from defects in material and workmanship for a specified			
	warranty period			
	25-2 The warranty period for compressors is 24 months from the			
	date of initial start-up or 30 months from the shipment date			
	of ex-work			
	START-UP & TRAINING FOR OPERATION & MAINTENANCE			
	26-1 Vendor shall be responsible for all start-up and complete			
the job for running machine				
26-2 Vendor shall offer the operational maintenance and				
26	training course for at least twenty-four hours			
	all design have to according future expansion			
	26-3 All design have to according future expansion			

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- END -